

MIELA MAYER • ESSAY

Questioning our Yield Gap

Agroforestry's Potential within the American Agriculture Industry

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A follow-on to *Learning to Let Nature Do the Work: A Review of Permaculture's Potential within the American Agriculture Industry* — extending that overview of the permaculture landscape into a closer look at agroforestry.

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Agroforestry's Potential within the American Agriculture Industry

Most Americans are unaware of the problems inherent in industrialized agriculture; however, the sector's heavy dependence on external, mechanized, and chemical inputs is exhausting the planet's finite supply of fossil fuels,^[1] arable cropland, and uncontaminated water, while directly contributing to greenhouse gas emissions and therefore overall climate change.^[2] According to the 2013 United Nations trade and environment review, agriculture, land use changes, and forestry account for a cumulative 71 percent of all greenhouse gas emissions.^[3] The American agriculture sector is primarily composed of monoculture-based systems that often have a prominence of genetically modified and herbicide resistant crops.^[4] Not only is this system unsustainable and environmentally damaging,^[5] but it is also now failing to provide the originally promised trade off for all its shortcomings: efficient and ample enough food production to keep up with the growing population's needs.^[6] Today we have 7 billion people looking to 5 billion hectares of farmland for their food, yet crop yields are only increasing at a rate of half a percent per year.^[7] Given that the world's population is expected to hit 9 billion people in 2050, this rate would need to more than quadruple in order to meet rising demands.^{[8] [9] [10]} On the bright side, while using the same science and technology that created "conventional farming techniques" people have also been able to design highly efficient, regenerative, and sustainable agriculture systems that are less reliant on fossil fuels and sequester more carbon.^[11] These methods, all of which fall under the overarching practice of permaculture, maintain healthy soil and support long-term productivity without synthetic chemical pesticides or fertilizers, thereby reducing petrochemical runoff into our rivers while using less water more efficiently.^{[12] [13]} The most modern subset of permaculture is agroforestry, and since it has the greatest documented potential to lower our carbon emissions, and therefore mitigate climate change, it will be the focus of this paper.

Similar to the green revolution's aim in shifting to industrial agriculture, agroforestry serves as a highly productive form of food production, requiring little external labor. The difference is that rather than working against natural processes and destroying the environment's pre-existing polycultures, agroforestry aims to constructively build upon such ecosystems, using the multipurpose diversity of species to construct a symbiotic, interdependent, and perennial supply of food.^{[14] [15]} The resulting resiliency of these sustainable methods, comparable to that of a natural forest, enable agroforestry to be applied and thrive virtually anywhere, unlike industrial farms; it is adaptable to a wide range of conditions specifically on a small scale.^[16] This means that over time widespread small scale agroforestry implementation is not only feasible but also provides the best means of reversing environmental degradation, preventing climate change through carbon sequestration, and reducing America's heavy reliance upon industrial agribusinesses.

Agroforestry: an Environmentally Beneficial Means to Close the American "Yield Gap" and Mitigate Climate Change

Our tendency is to hold back the environment's succession through weeding, mowing, pruning, etc. This "succession" however, is nature's sustainable way of cycling minerals and organic matter to efficiently produce food while maintaining soil ecology, promoting biotic diversity, preventing erosion, and providing habitats for other animal species—all without high labor inputs and fossil fuel consumption.^[17]

Agroforestry aims to mimic these naturally occurring processes creating a low input, self-sufficient, perennial edible forest. The idea of "forest gardening" is just about 40 years old and has only recently entered America.^[18] It was originally established by Robert Hart in Europe, where, with virtually no GMO's, and significantly less pesticide and fertilizer use, the European commodity crop yields have kept up with, and actually surpassed those of the United States.^[19] A study published by the *International Journal of Agricultural Sustainability* comparing the United Nations Food and Agriculture Organization's data on American and

European Maize production from 1961 to present, revealed that, in 1986 western Europe started producing slightly more than the US with 82899

hectograms/hectare and has remained in the lead ever since (see Figure 1 for trend data).

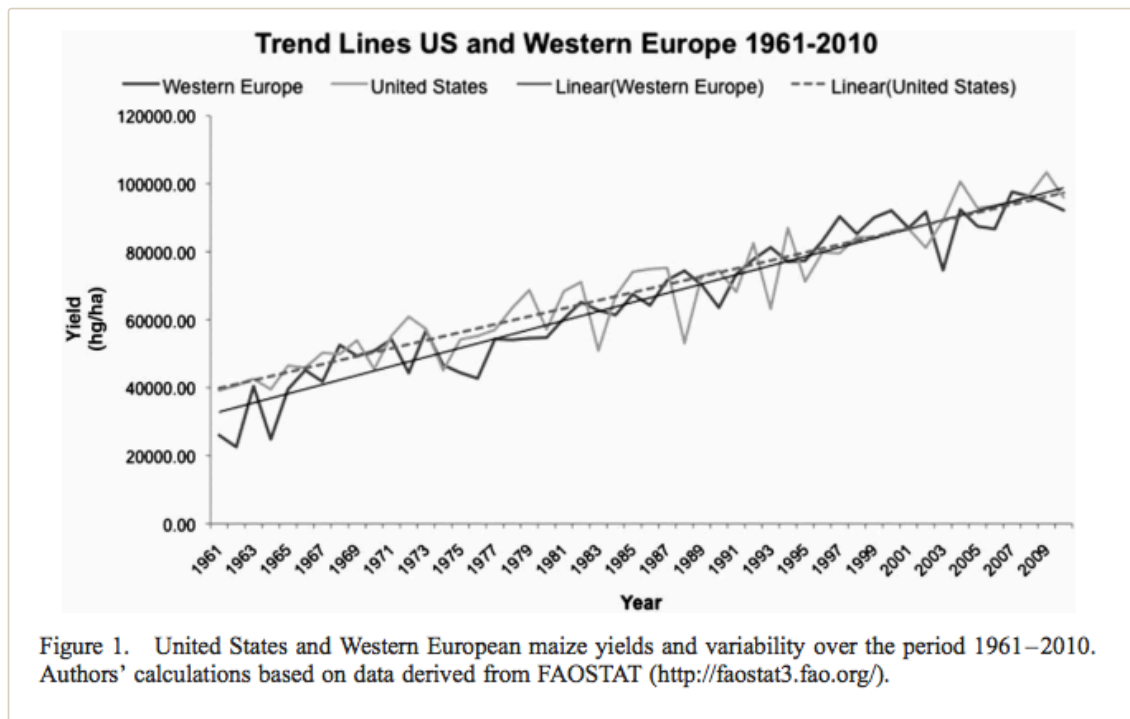


Figure 1: taken from Jack Heinemann's report: Sustainability and innovation in staple crop production in the US Midwest (2013)^[20]

Jack Heinemann, head of the study at the University of Canterbury in New Zealand, further points out that “the difference between the estimated yield potential and actual yield or ‘yield-gap’ appears to be uniformly smaller in Western Europe than in the US Midwest.”^[21] This means that we are not only producing less food than other more sustainable models, but we are using far more land and other resources to do so. Although it is hard to determine how widely used agroforestry practices are in Western Europe, these numbers at the very least suggest that alternative models for large-scale agriculture exist and are thriving without the vast majority of chemicals we in America commonly use, as many of these substances have been banned in Europe.

So how might agroforestry methods “double or triple crop yields while reducing the need for commercial fertilizers”?^[22] The principal focus is on creating a garden or farming system that mimics the structure and bene-

ficial relationships one would find in a naturally occurring ecosystem—essentially designing agriculture to look like a forest. By doing so, the food production site ends up becoming a diverse polyculture that integrates trees, shrubs, perennial and annual plants, to maintain the sustainable resilience and productivity of a natural forest.^[23] Agroforestry involves various processes, most of which fall under two main stages: remediating the soil using a technique called sheet mulching, and planting perennial polycultures to thrive within the healthier land. I will explain each of these stages in turn with regard to their potential to reverse environmental degradation, halt global warming, and decrease our reliance on a chemical and fossil fuel driven agriculture sector.

Agroforestry's Potential to Reverse Environmental Damage

Sheet Mulching:

Sheet mulching, the first step of agroforestry, is a process that returns organic matter, full of carbon and other vital nutrients to the soil, thereby rejuvenating

overused and degraded land, and enhancing its fertility. According to the 2011 FAO report, “25 percent of the world’s food-producing soils are highly degraded or are rapidly being degraded”; if we also take into account moderately degraded soils, “one-third of Earth’s entire endowment of cropland is under threat.”^[24] Sheet mulching is a cost effective and efficient way to transform any amount of unproductive or unused, degraded land into a fertile layer of carbon sequestering soil. Such regenerative processes are particularly important when working in an area that was once chemically treated.

Sheet mulching essentially involves four layers: the preexisting base (most often, brown field or a lawn), a nitrogen rich compost layer, some type of weed barrier, and a carbon rich cover mulch. The following sheet mulching diagram (Figure 2) was taken from a report on the University of Massachusetts’s on-campus permaculture start-up. Ryan Harb, head of the permaculture project, presented this diagram to show the four simple layers that he and fellow students used to remediate the soil on their 5000 square foot study site in 2009.

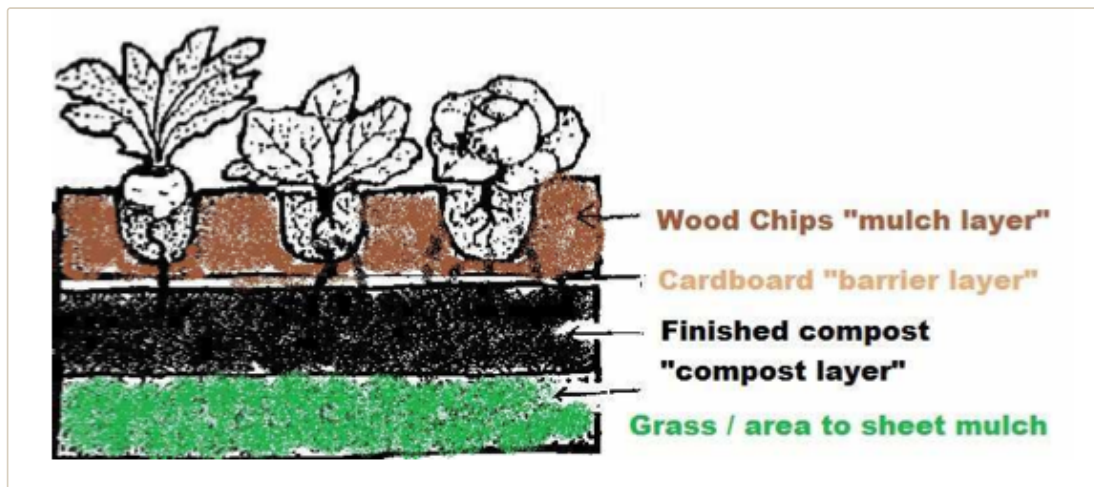


Figure 2: Sheet Mulching Diagram^[25]

According to Harb, each layer can be made out of different materials depending on what is locally available. The nitrogen rich layer is ideally composed of finished compost, but any mix of organic material like manure, slashed vegetation or food waste will break down over time and suffice. The entire base is covered in about three to four inches of this compost, but it can additionally be put on top of the weed barrier for extra nutrients. Cardboard or newspaper is usually the cheapest weed barrier and can be overlapped in multiple sheets to completely seal the compost. The last carbon rich layer, often woodchips, leaves, store bought mulch or straw, helps to insulate and protect everything. Sheet mulching can be done in the spring or fall, and is a good way to keep the land productive throughout the winter because the layers are just left to sit for at least 6 months. This allows the organic materials to circulate

and undergo natural processes to revive the soil without the use of chemical fertilizers.^[26]

Sheet mulching not only reduces the need for synthetic nutrient inputs (which can be costly) but also reduces the amount of herbicides and labor wasted on weeding. Weed formation often occurs when the land is too dry or loosely packed.^[27] These conditions require constant applications of micronutrient fertilizers in order to maintain fertile soil in combination with increasing amounts of herbicides to keep the weeds from taking over cropland. As Geoff Lawton, the managing director of The Permaculture Research Institute of Australia, explains, weeds function as a “volunteers” to help hold the soil together and prevent topsoil loss.^[28] If the process is successful in bringing the soil quality up to a standard level of health, we will have fewer weeds and therefore lower the amount of effort and chemicals necessary to support productive agriculture. Lowering

our reliance on such inputs has become especially critical in recent years considering the fact that “overall pesticide use increased by and estimated 183 million kgs (404 million pounds) between 1995 and 2011.”^[29] This increase contrasts harshly with Europe’s reduction of pesticide use to just 12% of 1995 levels during the same time period.^[30] From this perspective, the implementation of sheet mulching would profoundly mitigate environmental degradation by decreasing the amount of chemical runoff that pollutes water-sources.

With industrial agriculture methods now using up about 80% of the world’s fresh water and causing a 75% loss in crop diversity (and therefore ecological resilience), another key benefit of land regeneration is the soils improved water retention.^[31] According to Claire O’Connor, agriculture specialist at the Natural Resource Defense Council, “[e]ach 1 percent increase in soil organic matter helps the soil hold 20,000 gallons more water per acre.”^[32] Several other processes exist that do not directly fall under the practice of agroforestry, but have significant effects on the soil’s water capacity. See **appendix I** for how techniques such as Hugelkultur and land Keyline can be applied near the end of the sheet mulching cycle to further reverse environmental degradation.

Perennial Polycultures

The second phase of agroforestry is focused on intercropping different levels of vegetation to create a diverse and resilient agro-polyculture. The integration of perennials and annuals (a practice often referred to as Coppice^[33]) provides protection against soil erosion while also allowing for immediate productivity of seasonal crops within a perennial edible forest.^[34] It is important to take subterranean root zoning into account when placing different plants next to each other. By thoroughly studying the different root patterns of plant species and the different elements of the soil profile that they each require, we have the ability to strategically place plants so that they are mutually beneficial, rather than competing for the same nutrients at the same depth in the earth. Sustainable layouts like this allow for densely packed vegetation, and therefore higher food yields per unit of land. It is on this basis

that Geoff Lawton argues that we would need “about two to three percent of the present area that we use with industrial agriculture – two percent in equivalent area...to supply the world with the same nutrition that it presently requires, using permaculture [and therefore agroforestry] principles.”^[35] These edible forest gardens also result in a more natural niche that attracts ranges of bird and insect species—maintaining a full circle ecosystem that both feeds us, and helps to reverse our degradation of the biodiverse land.^[36]

The most manageable way to actually plan out where each plant will go is by selecting the tallest species to become a central element that the rest of the plant layers can be structured around. Depending on the size of the site, there will likely be more than one tall central species, each with multiple layers of diverse plants around it; these each become “guilds” or mutually beneficial miniature ecosystems, multiple of which make up an edible forest. Figure 5 presents a diagram of the multiple layers that should be present in each finalized guild. Certain plants play integral roles and benefit surrounding vegetation: soil improvers fix nitrogen levels, nectaries attract pollinating insects and prevent colony collapse, aromatic pest repellents and ground coverage shrubs lock nutrients in place while keeping unwanted vegetation out.^[37]

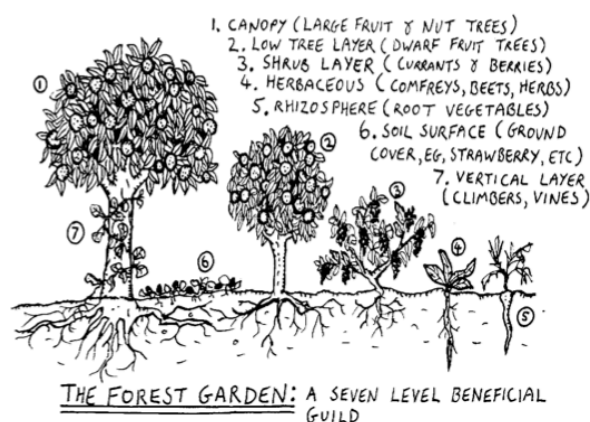


Figure 5: Structural layers of agroforestry^[38]

With our soil loss nearing about 75 billion tons annually, the key to any form of agroforestry is including perennials and ground covers.^[39] Perennial plants, which are often the larger woody species such as fruit and nut trees, have much more time to establish them-

selves and therefore larger subterranean root structures. These roots hold the topsoil in place while also keeping the dirt at its optimal compactness, which lessens the need for weeds.^[40] Ground covers that hold year round are beneficial for protecting vulnerable soils through out the winter, but this is not their only service. When left un-tilled, these perennial polycultures fosters a biodiverse array of flora and fauna in the soil by “providing organisms, such as earthworms, with food from the residues and by stabilizing their habitat.”^[41] Additionally, the remaining crop residues create a water filtering compost layer which limits runoff and acts as a buffer mitigating the amount of sediment, fertilizer and pesticide pollution entering nearby water sources.

“Perennial farming systems are particularly suited to stabilizing slopes and preventing erosion on hillside farms,” writes Eric Toensmire, the author of “Perennial Vegetables” and teaching fellow for carbon sequestering agroforestry at Yale University. This is important since “roughly 45% of the world's farmland is classed as sloping at 8% or higher—regeneration of this quantity of farmland with permanent agriculture would sequester 16.8 billion tons of carbon (at 25t/ha).”^[42] It is the woody nature of perennial roots, which make up about 80% of the plants biomass that allows them to sink such high concentrations of carbon in their bark.^[43] These roots are also more durable so they hold up better to insects and extreme weather conditions without requiring the same amount of maintenance and fertilizer as annuals.^[44] In fact, compared to conventional croplands, perennial permaculture methods that use agroforestry produced 24-34 percent greater yields during drought season while using 28-32 percent less energy.^[45] This difference in productivity comes from the superior water retention capacity of healthy soils containing a high prevalence of carbon and could be a contributing factor to Western Europe’s more narrow yield gap.

Agroforestry Techniques Mitigate Climate Change through Carbon Sequestration and Reduced Green House Gas Emissions

David R Higgins, soil scientist with the USDA-Agricultural Research Service Land Management and Water Conservation Research Unit, writes,

“One of agriculture’s main greenhouse gas mitigation strategies is soil carbon sequestration, wherein crops remove carbon dioxide from the atmosphere during photosynthesis, and non harvested residues and roots are converted to soil organic matter, which is 58 percent carbon.”^[46]

I have already discussed sheet mulching with regard to regenerating degraded soils by returning organic matter and micronutrients to the earth, but the resulting healthy soils also make up the largest terrestrial carbon reservoir within the biosphere, playing a significant role in capturing atmospheric CO₂.^[47] The healthier the soil, the greater capacity it’s microbes have to store atmospheric carbon. This storage intern helps to reduce ocean acidification, as the world’s oceans are saturated with approximately 3800 Pg (1 Pg = 1015g) and have limited ability to sink any more carbon without destroying its marine ecosystems.^[48] With global temperatures on the rise, and an estimated global loss of 50-70 percent of our soils original carbon content in just this past decade, it is crucial that we prioritize cultivating crops for optimal atmospheric carbon capture. See **Appendix II** for data on how much more carbon Europe has been able to sequester in their soil compared to that of the US agriculture sector. Agroforestry and sheet mulching practices seem to be the main differing factor between these two countries.

Studies show that “[c]onverting the 64 million hectares of U.S. cropland currently planted in corn and soybeans...would sequester 264 million tons of carbon dioxide; this is the equivalent of shutting down 207 (225- megawatt) coal-fired power plants, about 14 percent of the [America’s] installed coal electric capacity.”^[49] Not only would this conversion to sheet mulching be extremely helpful in mitigating climate change, but as students at UMass Amherst found out, the soil’s carbon rich content and overall ecology directly influences crop health and makes for much more flavorful produce. High quality fruit will have a high brix score, which is an indicator for carbon levels. The exact amount of carbon sequestered varies drastically

with humidity, location and implemented management. A properly designed full permaculture sight, that uses not only the fundamentals of agroforestry polycultures, but also other more traditional agroecological techniques that can be incorporated into to the sheet mulching process such as Hugelkultur,^{[50], [51]} Keyline,^[52] Coppice, and Biochar^{[53], [54]} is said to have an estimated potential of storing up to 228 tons (510720 pounds) of the world carbon in the soil.^[55] Furthermore, as Toensmire points out, about 1 million tons (200 million pounds)/year are already sequestered just through the relatively small amount of agroforestry that exists globally.

By nature of having a perennial edible forest, agroforestry practices do not till the land at any point. Conventional plow-based farming not only leaves soil vulnerable to erosion promoting agricultural runoff, but turning up the soil also releases all the carbon that had been stored below the surface. Perennial intercropping in agroforestry leads to a self-reliant more resilient guild that requires far less labor and assistance to thrive. These practices reduce a farmers need for machinery to spread pesticides, herbicides, or mechanically till the land and therefore uses 50 to 80% less fuel than conventional agriculture, directly mitigating green house gas emissions. In the 2013 United Nations Environment Program Report, Henry Neufeldt of the World Agroforestry Centre estimates that “no-tillage operations in the United States have helped avoid 241 million metric tons of carbon-dioxide.” Remarkably, all of this mitigation has occurred on just the dismal 10% of US crop lands dedicated to no till practices.^[56]

As discussed earlier perennial polycultures thrive on vulnerable sloping lands where they prevent topsoil loss and store carbon. A prevalent worry with these perennials is that they take too much time to establish and that they aren’t financially productive enough within the first year or two. In agroforestry however, the timing of productivity is species-dependent, and although the idea is to follow natural succession, we can actually accelerate this process. Eric Toensmire refers to this idea as “instant succession”. Instant succession forest gardens are ones that allow us to create all the naturally occurring levels of succession, essentially all

at once. Initially this method requires an intense investment of time and effort, but can result in a rapidly flourishing edible forest planted in dense patterns that should be self-sustaining without much maintenance for multiple years.^[57]

There is still much to be uncovered about which other plant species and techniques can be beneficial in creating an interdependent web of edible forest structures. “Over the next few years “farming” has the potential to be widely recognized as an intellectually rigorous and rewarding field ready for more exploration.”^[58] But for the spread of agroforestry to be possible we need much more research and recognition of the science behind carbon emissions and soil ecology. The social views around farming must shift to seeing it as an art of strategic planning and an intellectual science of designing ecosystems that mimic the structures in nature and maximize their interdependent benefits. If implemented on a large scale, agroforestry would be productive and resilient enough to eliminate the need for GMO crops, and result in massive cost cuts to the farmer.

Agroforestry’s Feasible Implementation on a Large Scale

Productivity

There is a belief that these pesticides, synthetic inputs, and modified seeds are the only way to feeding America, and that a system such as agroforestry could never reach the same productivity. However, the fact that the UK as surpassed growth while having banned many of the pesticides and chemicals we in America commonly use, suggests otherwise. The poor quality of our soil directly contributes to productivity loss and requires a constant flow of chemical inputs just to be able to support plant life.^[59] It turns out, herbicide-resistant crop technologies such as seeds genetically modified to be “Roundup Ready”, have actually “led to a 239 million kilogram (527 million pound) increase in herbicide use in the United States between 1996 and 2011.”^[60] Furthermore, farmers are now struggling with a new emergence of HT resistant “super-weeds.” Clearly GM technologies are only furthering our pesticide burden and creating entirely new problems.

Agroforestry systems on the other hand, can actually approach the same plant density as many cash crops genetically modified to have their weed structures grow directly downwards for tight packing in the field.^[61] We can do this by taking into account each species subterranean structure and strategically planting to ensure that the roots of one plant do not compete with adjacent vegetation.

Resiliency

UN secretariat Ulrich Hoffman, in reference to our current agriculture paradigm, has stressed that: “As the world currently follows the worst case scenario of green house gas emissions projected by Intergovernmental Panel on Climate Change and International Energy Agency (i.e. implying a global warming of 4-6 degrees in Celsius) climate resilience and adaption, in combination with productivity increases should be prioritized.”^[62] Agroforestry not only provides a feasible option for narrowing our yield gap as demonstrated by Europe, but it is also a good safety plan as we do begin to face the effects of climate change. Healthier soils full of organic matter are naturally more resilient to droughts and floods as they become more “sponge like” and therefore durable to the effects of climate change.^[63] With self sustaining crops comes immense cuts in pesticide input costs making agroforestry actually more cost effective. As Gabe Brown, owner of a 5,000-acre farm in North Dakota, explains, with less labor and lower costs, his farm no longer requires nitrogen fertilizers or fungicides to produce yields that regularly top the countries average. According to Mr. Brown, these techniques will gain converts “as growers increasingly face extreme weather, high production costs, a shortage of labor and the threat of government regulation of agricultural pollution.”^[64] The natural resiliency of producing in edible forests could also cut costs by negating the need for the “safety net” farmer subsidies currently built into our farm bill.

Why small scale is the best place to start lowering our reliance on industrial agribusinesses

At the moment agroforestry is by no means a replacement for all the commercial fields of commodity crops,

but unlike conventional farming, these techniques can be applied almost anywhere, on small and large scales.

^[65] This means that any patch of unproductive grass can be converted into a thriving, food producing micro ecosystem. Such forest gardens could be in every park, making up the front yards of churches, town halls, and eventually individual homes. Even though in the long run Agroforestry practices would not just be feasible, but beneficial on a larger scale, in the short term drastic steps in this direction are unlikely due to general infrastructural challenges, as all our machinery, laws, and subsidies are all structured around a completely different system. So for now I propose that we direct attention to implementing agroforestry practices, or at the very least some form of sheet mulching to sequester carbon, on a small scale in as many locations as possible. The most recent Farm Bill was just renewed in 2014 and won't be up for policy review for approximately another 5 years. This means that we have five years during which we can begin to shift our reliance from BigAg to more local carbon sequestering and environmentally regenerative food production sites. Five years is certainly enough time for a movement to take get off the ground. As time goes on and more people are exposed it will become much easier to lobby for more and more infrastructural changes to start implementing such plans on a larger scale. Implementing agroforestry practices on a small scale is very straight forward, cost effective, and only labor intensive during the early sheet mulching stages. As Toensmire puts it, “Permanent agriculture needs to be spread much, much more widely if it is to have a significant impact.”^[66] Sustainable food production through agroforestry techniques provides a means for nature to reclaim lost land one patch of chemically treated, mechanically watered, constantly mowed grass at a time—starting on a small scale and spreading all over both urban and rural locations. Eventually the resulting “bottom up” approach to changing our food system makes each community more locally self-sufficient with the potential to lessen America's reliance on large-scale unsustainable agribusinesses. But the first, and much bigger step for America is recognizing that land is lost; that just because we see green grass, use of fertilizer on it, and have a supermarket full of food, does not mean

our system is at all flawless. We have to overcome this common inertia towards pressing environmental issues and begin to implement existing knowledge.

Index I

Hugelkultur and Land Keyline

Hugelkultur and land Keyline, can be done during the sheet mulching cycle. These are both labor-intensive steps but they can result in more productive returns from the land. Hugelkultur is a means of minimizing water use and maximizing the land's water retention. It is a great option especially in cold climates because it extends the productive season of any site. Hugelkultur involves burying wood, full of carbon and other organic matter, deep below the ground.^[67] This wood is used to improve the earth's water retention capacity and will break down over time furthering the soil's biodiversity. To make it less work intensive one can also make "

Hugelburms" which are basically raised flower beds made from simply stacking the wood on of the exposed ground layer and then preforming normal sheet mulching over them.^[68] This technique works just as well in terms of water retention, and also has the added benefit of enabling control over the shape of the land—a Keyline technique. Keyline farming is based on altering the contour line that forms with the slope of the land in order to adjust wind patterns and increase topsoil.^[69] The combination of Hugelkultur and Keyline farming allow for the most efficient use of water and the ability to slow water flow, maximizing its coverage by redirecting the flow of gravity all over the permaculture sight.^[70] This practice has most recently been used by Ben Falk, a homestead permaculture farmer in central VT, who integrated and used both Keyline and

Hugelkultur to alter the contour of the land and work with the wind patterns.^[71]

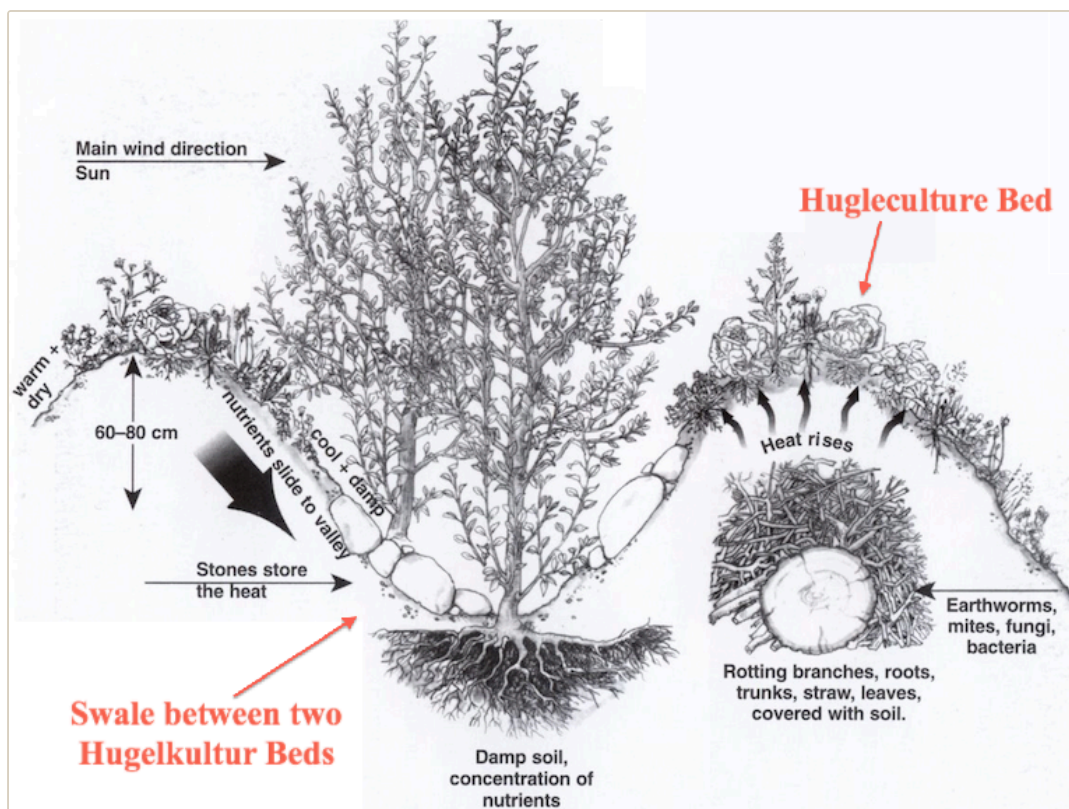


Figure 4: Hugelkultur and Keyline working together^[72]

Figure 4 illustrates how Keyline farming can use two Hugelburms to create a swale in between where the wind is blocked and the ecosystem feels similar to that

of a valley. Here and plants can flourish at high altitudes in temperate zones for a longer, moister season than usual.

Index II

Comparing data in the United Kingdom to the United States

Our topsoil is more eroded and has significantly less carbon content storage than in the UK. In America we

also emit more gigagrams of CO2 from agricultural energy use and apply more pesticides on average.

Soil Erosion:

United Kingdom	Average soil erosion expressed in GLASOD erosion degree	Soil	1991	degrees	0.77		
United States of America	Average soil erosion expressed in GLASOD erosion degree	Soil	1991	degrees	0.87		

Carbon Content:

United States of America	Soil	Average carbon content in the topsoil as a % in weight [%]	1.52
United Kingdom	Soil	Average carbon content in the topsoil as a % in weight [%]	6.98

Agricultural emissions from energy use:

Emissions of carbon dioxide, methane and nitrous oxide gases associated with fuel burning and electricity generation in agriculture.

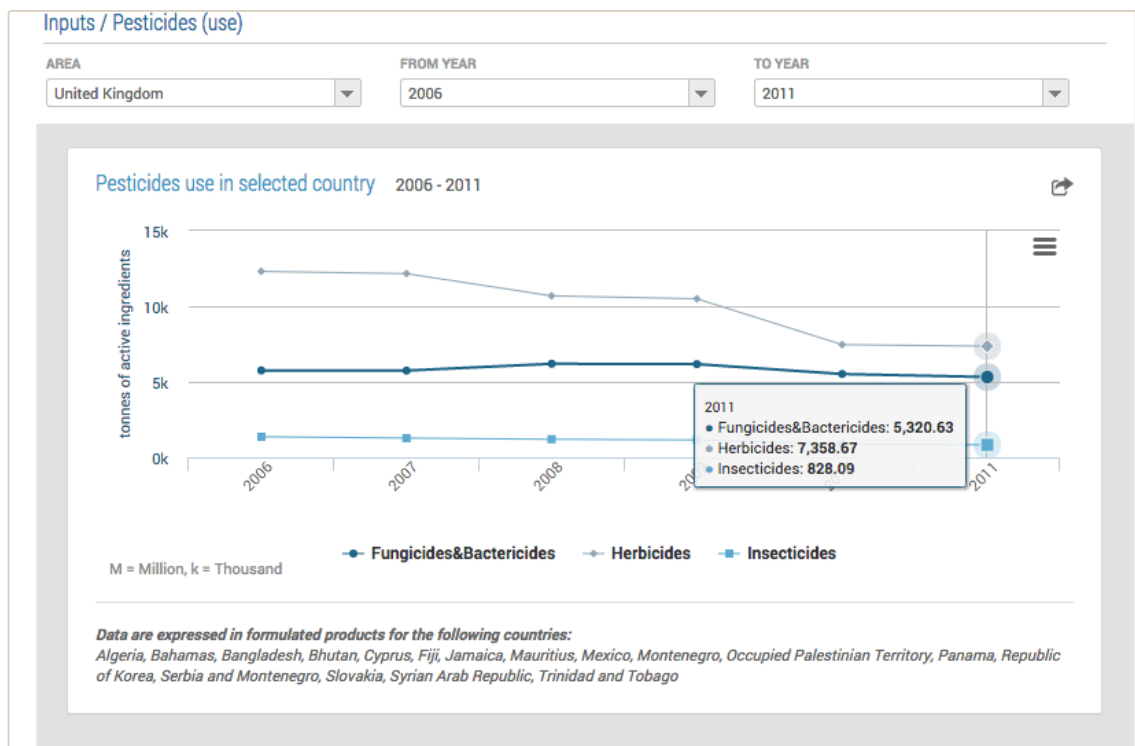
Agriculture Total	United Kingdom	Emissions (CO2eq)	Energy Use	2011	Gigagrams	3150.14		
Agriculture Total	United States of America	Emissions (CO2eq)	Energy Use	2011	Gigagrams	48040.62		

Source:

and

<http://data.fao.org/dataset-data-filter?entryId=a51018a8-6b05-4694-92ff-26e26c732c70&tab=data&type=Measure&uuidResource=fd1ee060-9eb8-4b39-bf25-0bcac38ad597>

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Comparing pesticide inputs graphically
Inputs/pesticide use in the UK



Inputs/pesticide use in the US



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 54. *This process involves adding an additional layer of recycled charcoal into the mix because it retains huge amounts of carbon even years after burning. It is important to note however, that such use of charcoal is only beneficial if it is from a recycled source, seeing as it would waste just as much carbon to intentionally burn*

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